

TAGORE COLLEGE OF ARTS AND SCIENCE

**(Affiliated to University of Madras)
Chrompet, Chennai-44**



DEPARTMENT OF COMPUTER SCIENCE

RELATIONAL DATABASE MANAGEMENT SYSTEM

NAME :

CLASS :

REG NO :

TAGORE COLLEGE OF ARTS AND SCIENCE

(Affiliated to the University of Madras)

Chrompet, Chennai – 600 044



DEPARTMENT OF COMPUTER SCIENCE

Certified to be the Bonafide Record of Practical work done by Mr. /Miss.
..... Reg. No..... of Third Year B.Sc., in
the Department of Computer Science, Shift-II, Tagore College of Arts & Science,
Chrompet, Chennai – 600 044 during the year 2025-2026

STAFF IN-CHARGE

HEAD OF THE DEPARTMENT

Submitted for B.Sc., Degree Practical Examination held on
at the Department of Computer Science, Shift-II, Tagore College of Arts and
Science, Chrompet, Chennai –44.

Date :

Examiners

1.

2.

INDEX

S. No	DATE	EXERCISE	Page No	Sign
1.		DDL COMMENTS WITH CONSTRAINTS		
2.		DML COMMENTS WITH CONSTRAINTS		
3.		QUERIES, SUBQUERIES & AGGREGATE FUNCTION		
4.		PL/SQL – EXECUTION HANDLING		
5.		PL/SQL – CURSOR		
6.		PL/SQL – TRIGGER		
7.		PL/SQL – PACKAGES		
8.		LIBRARY MANAGEMENT SYSTEM		
9.		STUDENT MARKSHEET PROCESSING		
10.		PAYROLL PROCESSING		

1) DDL COMMENTS WITH CONSTRAINTS

DDL commands:

- * CREATE
- * ALTER
- * DROP
- * RENAME
- * TRUNCATE

Queries:

CREATE TABLE

Q1 → create table student(stid number, sname varchar(20), score integer);

Q2 → desc student;

Q3 → create table stu(sid number primary key, sname varchar(20) not null, email varchar(30) unique, age decimal(10,2) check (age>=18));

Q4 → desc stu;

ALTER TABLE

Q5 → alter table student drop column score;

Q6 → alter table student add gender varchar(5);

Q7 → alter table stu modify age number;

RENAME TABLE

Q8 → rename student to stu10;

Q9 → desc stu10;

Q10 → alter table stu rename column email to emailid;

Q11 → desc stu;

TRUNCATE AND DROP

Q12 → truncate table stu;

Q13 → drop table stu;

Q14 → desc stu;

OUTPUT:

CREATE TABLE

O/P → Table created.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STUDENT</u>	<u>STID</u>	Number	-	-	-	-	✓	-	-
	<u>STNAME</u>	Varchar2	20	-	-	-	✓	-	-
	<u>SCORE</u>	Number	-	-	0	-	✓	-	-

O/P →

O/P → Table created.

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STU</u>	<u>SID</u>	Number	-	-	-	1	-	-	-
	<u>SNAME</u>	Varchar2	20	-	-	-	-	-	-
	<u>EMAIL</u>	Varchar2	30	-	-	-	✓	-	-
	<u>AGE</u>	Number	-	10	2	-	✓	-	-

ALTER TABLE

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STUDENT</u>	<u>STID</u>	Number	-	-	-	-	✓	-	-
	<u>STNAME</u>	Varchar2	20	-	-	-	✓	-	-

O/P → Table altered.

O/P → Table altered.

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STU</u>	<u>SID</u>	Number	-	-	-	1	-	-	-
	<u>SNAME</u>	Varchar2	20	-	-	-	-	-	-
	<u>EMAIL</u>	Varchar2	30	-	-	-	✓	-	-
	<u>AGE</u>	Number	-	-	-	-	✓	-	-

RENAME TABLE

O/P → Statement processed.

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STU10</u>	<u>STID</u>	Number	-	-	-	-	✓	-	-
	<u>STNAME</u>	Varchar2	20	-	-	-	✓	-	-

	<u>GENDER</u>	Varchar2	5	-	-	-	✓	-	-
--	-------------------------------	----------	---	---	---	---	---	---	---

O/P →

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
<u>STU</u>	<u>SID</u>	Number	-	-	-	1	-	-	-
	<u>SNAME</u>	Varchar2	20	-	-	-	-	-	-
	<u>EMAILID</u>	Varchar2	30	-	-	-	✓	-	-
	<u>AGE</u>	Number	-	-	-	-	✓	-	-

TRUNCATE AND DROP

O/P → Table truncated.

O/P → Table dropped.

O/P → Object to be described could not be found.

2) DML COMMENTS WITH CONSTRAINTS

DML commands:

- * INSERT
- * UPDATE
- * DELETE
- * SELECT

Queries:

INSERT QUERY

Q1 → insert into stu10(stid,stname,gender) values (1001,'aaa','f');

Q2 → insert into stu10(stid,stname,gender) values (1002,'bbb','f');

Q3 → insert into stu10(stid,stname,gender) values (1003,'cccc','f');

Q4 → insert into stu10(stid,stname,gender) values (1004,'ddd','m');

Q5 → insert into stu10(stid,stname,gender) values (1005,'EEE','M');

SELECT QUERY

Q6 → select * from stu10;

Q7 → select stname from stu10;

Q8 → select * from stu10 where gender='f';

UPDATE QUERY

Q9 → update stu10 set stname='anu' where stid=1001;

Q10 → update stu10 set stname='bala', gender='m' where stid=1002;

Q11 → select * from stu10;

DELETE QUERY

Q12 → delete from stu10 where stid=1003;

Q13 → select * from stu10;

Q14 → delete from stu10;

Q15 → select * from stu10;

OUTPUT:

INSERT QUERY

O/P → 1 row(s) inserted.

SELECT QUERY

O/P →

STID	STNAME	GENDER
1001	aaa	f
1002	bbb	f
1003	cccc	f
1004	ddd	m
1005	EEE	M

O/P →

STNAME
aaa
bbb
cccc
ddd
EEE

O/P →

STID	STNAME	GENDER
1001	aaa	f

1002	bbb	f
1003	cccc	f

UPDATE QUERY

O/P → 1 row(s) updated.

O/P → 1 row(s) updated.

O/P →

STID	STNAME	GENDER
1001	anu	f
1002	bala	m
1003	cccc	f
1004	ddd	m
1005	EEE	M

DELETE QUERY

O/P → 1 row(s) deleted.

O/P →

STID	STNAME	GENDER
1001	anu	f
1002	bala	m
1004	ddd	m

1005	EEE	M
------	-----	---

O/P → 4 row(s) deleted.

O/P → no data found

3) QUERIES, SUBQUERIES AND AGGREGATE FUNCTION

Queries:

BASIC QUERY

Q1 → create table empl(eid number primary key, ename varchar(20), deptid number, deptname varchar(20), salary number);

Q2 → insert into empl(eid,ename,deptid,deptname,salary) values (1201,'Arun',104,'CS',50000);

Q3 → insert into empl(eid,ename,deptid,deptname,salary) values (1202,'Bala',104,'CS',60000);

Q4 → insert into empl(eid,ename,deptid,deptname,salary) values (1203,'Chandru',105,'COMMERCE',40000);

Q5 → insert into empl(eid,ename,deptid,deptname,salary) values (1204,'Dilip',104,'CS',45000);

Q6 → insert into empl(eid,ename,deptid,deptname,salary) values (1205,'Dhanu',106,'TAMIL',30000);

Q7 → insert into empl(eid,ename,deptid,deptname,salary) values (1206,'Guru',106,'TAMIL',58000);

Q8 → insert into empl(eid,ename,deptid,deptname,salary) values (1207,'Hari',105,'COMMERCE',38000);

SELECT QUERY

Q9 → select * from empl;

Q10 → select eid,ename,salary from empl;

Q11 → select * from empl where salary >= 50000;

AGGREGATE FUNCTION

Q12 → select COUNT(*) as totalemp from empl;

Q13 → select SUM(salary) as totalsal from empl;

Q14 → select AVG(salary) as totalsal from empl;

Q15 → select MAX(salary) as maxsalary from empl;

Q16 → select MIN(salary) as minsalary from empl;

Q17 → select eid,salary from empl order by salary desc;

Q18 → select deptid, sum(salary) as sal from empl group by deptid;

SUBQUERIES

Q19 → select * from empl where salary > (select avg(salary) from empl);

Q20 → create table dept(deptid number,deptname varchar(20));

Q21 → insert into dept(deptid,deptname) values (104,'CS');

Q22 →insert into dept(deptid,deptname) values (105,'COMMERCE');

Q23 →insert into dept(deptid,deptname) values (106,'TAMIL');

Q24 →insert into dept(deptid,deptname) values (107,'ENGLISH');

Q25 →select * from dept;

--- List department wise employee count:

Q26 →select d.deptname, e.empcount from dept d JOIN (select deptid, count(*) as empcount from empl group by deptid) e on d.deptid = e.deptid;

JOIN QUERIES

Q27 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e join dept d on e.deptid = d.deptid;

Q28 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e LEFT JOIN dept d on e.deptid = d.deptid;

Q29 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e RIGHT JOIN dept d on e.deptid = d.deptid;

Q30 →select e.eid, e.ename, e.salary, d.deptid, d.deptname from empl e CROSS JOIN dept d;

OUTPUT:

BASIC QUERY

O/P → Table created.

SELECT QUERY

O/P →

EID	ENAME	DEPTID	DEPTNAME	SALARY
1201	Arun	104	CS	50000
1202	Bala	104	CS	60000
1203	Chandru	105	COMMERCE	40000
1204	Dilip	104	CS	45000
1205	Dhanu	106	TAMIL	30000
1206	Guru	106	TAMIL	58000
1207	Hari	105	COMMERCE	38000

O/P →

EID	ENAME	SALARY
1201	Arun	50000
1202	Bala	60000
1203	Chandru	40000
1204	Dilip	45000
1205	Dhanu	30000
1206	Guru	58000
1207	Hari	38000

O/P →

EID	ENAME	DEPTID	DEPTNAME	SALARY
1201	Arun	104	CS	50000
1202	Bala	104	CS	60000
1206	Guru	106	TAMIL	58000

AGGREGATE FUNCTION

O/P →

TOTALEMP
7

O/P →

TOTALSAL
321000

O/P →

TOTALSAL
45857.1428571428571428571428571429

O/P →

MAXSALARY
60000

O/P →

MINSALARY
30000

O/P →

EID	SALARY
1202	60000
1206	58000
1201	50000
1204	45000
1203	40000
1207	38000
1205	30000

O/P →

DEPTID	SAL
104	155000
105	78000
106	88000

SUBQUERIES

O/P →

EID	ENAME	DEPTID	DEPTNAME	SALARY
------------	--------------	---------------	-----------------	---------------

1201	Arun	104	CS	50000
1202	Bala	104	CS	60000
1206	Guru	106	TAMIL	58000

O/P →

DEPTID	DEPTNAME
104	CS
105	COMMERCE
106	TAMIL
107	ENGLISH

O/P →

DEPTNAME	EMPCOUNT
COMMERCE	2
CS	3
TAMIL	2

JOIN QUERIES

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	105	COMMERCE

1204	Dilip	45000	104	CS
1205	Dhanu	30000	106	TAMIL
1206	Guru	58000	106	TAMIL
1207	Hari	38000	105	COMMERCE

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1204	Dilip	45000	104	CS
1202	Bala	60000	104	CS
1201	Arun	50000	104	CS
1207	Hari	38000	105	COMMERCE
1203	Chandru	40000	105	COMMERCE
1206	Guru	58000	106	TAMIL
1205	Dhanu	30000	106	TAMIL

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	105	COMMERCE
1204	Dilip	45000	104	CS
1205	Dhanu	30000	106	TAMIL
1206	Guru	58000	106	TAMIL

1207	Hari	38000	105	COMMERCE
-	-	-	107	ENGLISH

O/P →

EID	ENAME	SALARY	DEPTID	DEPTNAME
1201	Arun	50000	104	CS
1202	Bala	60000	104	CS
1203	Chandru	40000	104	CS
1204	Dilip	45000	104	CS
1205	Dhanu	30000	104	CS
1206	Guru	58000	104	CS
1207	Hari	38000	104	CS
1201	Arun	50000	105	COMMERCE
1202	Bala	60000	105	COMMERCE
1203	Chandru	40000	105	COMMERCE

4) PL/SQL – EXCEPTION HANDLING

Queries:

Predefined Exceptions

Q1 →

DECLARE

num1 NUMBER := 10;

num2 NUMBER := 0;

result NUMBER;

BEGIN

result := num1 / num2;

DBMS_OUTPUT.PUT_LINE('Result: ' || result);

EXCEPTION

WHEN ZERO_DIVIDE THEN

DBMS_OUTPUT.PUT_LINE('Cannot divide by zero.');

END;

User-Defined Exceptions

Q2 →

DECLARE

insufficient_balance EXCEPTION;

balance NUMBER := 500;

BEGIN

IF balance < 1000 THEN

RAISE insufficient_balance;

END IF;

EXCEPTION

WHEN insufficient_balance THEN

```
DBMS_OUTPUT.PUT_LINE('Balance is too low.');
```

END;

Named System Exceptions

Q3 →

```
DECLARE
    ename emp.name%TYPE;
BEGIN
    SELECT name into ename
    FROM emp
    WHERE eid = 1001; -- assuming invalid ID
    DBMS_OUTPUT.PUT_LINE('Employee Name: ' || ename);
EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No employee found with that ID.');
```

END;

Unhandled (OTHERS) Exception

Q4 →

```
BEGIN
    EXECUTE IMMEDIATE 'DROP TABLE nonexistent_table';
EXCEPTION
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Some unexpected error occurred: ' || SQLERRM);
END;
```

OUTPUT:

Predefined Exceptions

O/P →

Cannot divide by zero.

User-Defined Exceptions

O/P →

Balance is too low.

Named System Exceptions

O/P →

No employee found with that ID.

Unhandled (OTHERS) Exception

O/P →

Some unexpected error occurred: ORA-00942: table or view does not exist

5) PL/SQL – CURSOR

Queries:

Q1 → select * from emp;

Implicit Cursor

Q2 →

BEGIN

UPDATE emp SET salary = salary + 1000 WHERE eid = 103;

IF SQL%ROWCOUNT = 1 THEN

DBMS_OUTPUT.PUT_LINE('1 row updated successfully.');

END IF;

END;

Q3 → select * from emp;

Explicit Cursor

Q4 →

DECLARE

CURSOR emp_cursor IS SELECT eid, name FROM emp;

v_id emp.eid%TYPE;

v_name emp.name%TYPE;

BEGIN

OPEN emp_cursor;

LOOP

FETCH emp_cursor INTO v_id, v_name;

EXIT WHEN emp_cursor%NOTFOUND;

DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Name: ' || v_name);

```
END LOOP;

CLOSE emp_cursor;

END;
```

Cursor FOR Loop

Q5 →

```
BEGIN

    FOR emp_rec IN (SELECT eid, name FROM emp) LOOP

        DBMS_OUTPUT.PUT_LINE('ID: ' || emp_rec.eid || ', Name: ' || emp_rec.name);

    END LOOP;

END;
```

Parameterized Cursor

Q6 →

```
DECLARE

    CURSOR emp_cursor(p_dept_id NUMBER) IS

        SELECT eid, name FROM emp WHERE dept_id = p_dept_id;

    v_id emp.eid%TYPE;

    v_name emp.name%TYPE;

BEGIN

    OPEN emp_cursor(123); -- passing department id

    LOOP

        FETCH emp_cursor INTO v_id, v_name;

        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE('ID: ' || v_id || ', Name: ' || v_name);

    END LOOP;

    CLOSE emp_cursor;

END;
```

OUTPUT:

O/P →

EID	NAME	DOJ	SALARY	DEPT_ID
103	Cholan	12-FEB-20	30000	123
104	Hari	22-FEB-24	20000	122
101	Arun	12-FEB-20	40000	123
102	Bala	01-DEC-19	50000	121
105	Guru	10-JAN-18	60000	122
106	Mathi	01-JUN-20	46000	121

Implicit Cursor

O/P →

1 row updated successfully.

1 row(s) updated.

O/P →

EID	NAME	DOJ	SALARY	DEPT_ID
103	Cholan	12-FEB-20	31000	123
104	Hari	22-FEB-24	20000	122
101	Arun	12-FEB-20	40000	123
102	Bala	01-DEC-19	50000	121
105	Guru	10-JAN-18	60000	122
106	Mathi	01-JUN-20	46000	121

Explicit Cursor

O/P →

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

Cursor FOR Loop

O/P →

ID: 103, Name: Cholan

ID: 104, Name: Hari

ID: 101, Name: Arun

ID: 102, Name: Bala

ID: 105, Name: Guru

ID: 106, Name: Mathi

Parameterized Cursor

O/P →

ID: 103, Name: Cholan

ID: 101, Name: Arun

6) PL/SQL – TRIGGER

PL/SQL Queries:

Q1 → CREATE TABLE students (roll_no NUMBER PRIMARY KEY,
name VARCHAR2(50), marks NUMBER);

BEFORE INSERT TRIGGER

Q2 →

```
CREATE OR REPLACE TRIGGER trg_before_insert_students
BEFORE INSERT ON students
FOR EACH ROW
BEGIN
    -- If marks not given, set to 0
    IF :NEW.marks IS NULL THEN
        :NEW.marks := 0;
    END IF;
END;
```

AFTER INSERT TRIGGER

Q3 →

```
CREATE OR REPLACE TRIGGER trg_after_insert_students
AFTER INSERT ON students
BEGIN
    DBMS_OUTPUT.PUT_LINE('A new student was added!');
END;
```

Q4 → INSERT INTO students (roll_no, name) VALUES (1, 'John');

Q5 → select * from students;

BEFORE UPDATE TRIGGER

Q6 →

```
CREATE OR REPLACE TRIGGER trg_before_update_students
BEFORE UPDATE ON students
FOR EACH ROW
BEGIN
    -- Prevent marks from being more than 100
    IF :NEW.marks > 100 THEN
        :NEW.marks := 100;
    END IF;
END;
```

Q7 → UPDATE students SET marks = 120 WHERE roll_no = 1;

Q8 → select * from students;

AFTER DELETE TRIGGER

Q9 →

```
CREATE OR REPLACE TRIGGER trg_after_delete_students
AFTER DELETE ON students
BEGIN
    DBMS_OUTPUT.PUT_LINE('A student record was deleted!');
END;
```

Q10 → DELETE FROM students WHERE roll_no = 1;

OUTPUT:

O/P →

Table Created.

BEFORE INSERT TRIGGER

O/P →

Trigger created.

AFTER INSERT TRIGGER

O/P →

Trigger created.

O/P →

A new student was added!

1 row(s) inserted.

O/P →

ROLL_NO	NAME	MARKS
1	John	0

BEFORE UPDATE TRIGGER

O/P →

Trigger created.

O/P →

1 row(s) updated.

O/P →

ROLL_NO	NAME	MARKS
1	John	100

AFTER DELETE TRIGGER

O/P → Trigger created.

O/P → A student record was deleted!

1 row(s) deleted.

7) PL/SQL – PACKAGES

PL/SQL Queries:

Package Specification

Q1 →

```
CREATE OR REPLACE PACKAGE math_pkg AS  
    FUNCTION square_num(p_num NUMBER) RETURN NUMBER;  
END math_pkg;
```

Package Body

Q2 →

```
CREATE OR REPLACE PACKAGE BODY math_pkg AS  
    FUNCTION square_num(p_num NUMBER) RETURN NUMBER IS  
    BEGIN  
        RETURN p_num * p_num;  
    END;  
END math_pkg;
```

Test the Package

Q3 →

```
BEGIN  
    DBMS_OUTPUT.PUT_LINE('Square of 5: ' || math_pkg.square_num(5));  
END;
```

Predefined Package

Q4 →

```
BEGIN  
    -- Random number between 1 and 50  
    DBMS_OUTPUT.PUT_LINE
```

```
('Random Number: ' || ROUND(DBMS_RANDOM.VALUE(1, 50)));
```

```
-- Random uppercase string of length 8
```

```
DBMS_OUTPUT.PUT_LINE('Random String: ' || DBMS_RANDOM.STRING('U', 8));
```

```
END;
```

OUTPUT:

Package Specification

O/P →

Package created.

Package Body

O/P →

Package Body created.

Test the Package

O/P →

Square of 5: 25

Predefined Package

O/P →

Random Number: 34

Random String: ZOWXBLHK

8) LIBRARY MANAGEMENT SYSTEM

Coding:

FORM1

Private Sub addcmd_Click()

form2.Show

End Sub

Private Sub issuecmd_Click()

form3.Show

End Sub

Private Sub membercmd_Click()

Form5.Show

End Sub

Private Sub reportcmd_Click()

DataReport1.Show

End Sub

Private Sub returncmd_Click()

form4.Show

End Sub

FORM 2

Dim cn As New ADODB.Connection

Dim rs As New ADODB.Recordset

Private Sub addcmd_Click()

rs.AddNew

```
rs.Fields(0).Value = Text1.Text
rs.Fields(1).Value = Text2.Text
rs.Fields(2).Value = Text3.Text
rs.Fields(3).Value = Text4.Text
rs.Update
```

```
MsgBox "Succssfully Inserted"
```

```
Text1.Text = ""
```

```
Text2.Text = ""
```

```
Text3.Text = ""
```

```
Text4.Text = ""
```

```
End Sub
```

```
Private Sub closecmd_Click()
```

```
form1.Show
```

```
End Sub
```

```
Private Sub Form_Load()
```

```
cn.Open "Provider=Microsoft.Jet.OLEDB.4.0;Data Source=E:\lib.mdb;Persist Security  
Info=False"
```

```
rs.Open "select * from book", cn, adOpenKeyset, adLockOptimistic
```

```
End Sub
```

FORM 3

```
Dim cn As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

```
Private Sub Command2_Click()
```

```
form1.Show
```

```
End Sub
```

```
Private Sub Form_Load()
```

```

cn.Open "Provider=Microsoft.Jet.OLEDB.4.0;Data Source=E:\lib.mdb;Persist Security
Info=False"
rs.Open "select * from book", cn, adOpenKeyset, adLockOptimistic
Text1.Enabled = False
Text2.Enabled = False
End Sub

```

Private Sub issuecmd_Click()

```

Dim bid As String
Dim sql, sql1 As String
Set rs = New ADODB.Recordset
bid = InputBox("Enter the Book ID :")
sql = "SELECT * FROM book WHERE bookid = " & bid
rs.Open sql, cn, adOpenForwardOnly, adLockReadOnly

```

```

If rs.EOF Then
    MsgBox "No record found with ID: "
    rs.Close
    Exit Sub
End If

```

```

Text1.Text = rs.Fields(0).Value
Text2.Text = rs.Fields(1).Value
sql1 = "update book set quan= quan-1 where bookid= " & bid
cn.Execute sql1
MsgBox "Book Issued"
End Sub

```

FORM 4

```

Dim cn As New ADODB.Connection
Dim rs As New ADODB.Recordset
Private Sub Command2_Click()
form1.Show

```

End Sub

Private Sub Form_Load()

```
cn.Open "Provider=Microsoft.Jet.OLEDB.4.0;Data Source=E:\lib.mdb;Persist Security  
Info=False"
```

```
rs.Open "select * from book", cn, adOpenKeyset, adLockOptimistic
```

```
Text1.Enabled = False
```

```
Text2.Enabled = False
```

End Sub

Private Sub retcmd_Click()

```
Dim bid As String
```

```
Dim sql, sql1 As String
```

```
Set rs = New ADODB.Recordset
```

```
bid = InputBox("Enter the Book ID to return book :")
```

```
sql = "SELECT * FROM book WHERE bookid = " & bid
```

```
rs.Open sql, cn, adOpenForwardOnly, adLockReadOnly
```

```
If rs.EOF Then
```

```
MsgBox "No record found with ID: "
```

```
rs.Close
```

```
Exit Sub
```

```
End If
```

```
Text1.Text = rs.Fields(0).Value
```

```
Text2.Text = rs.Fields(1).Value
```

```
sql1 = "update book set quan= quan+1 where bookid= " & bid
```

```
cn.Execute sql
```

```
MsgBox "Book Returned"
```

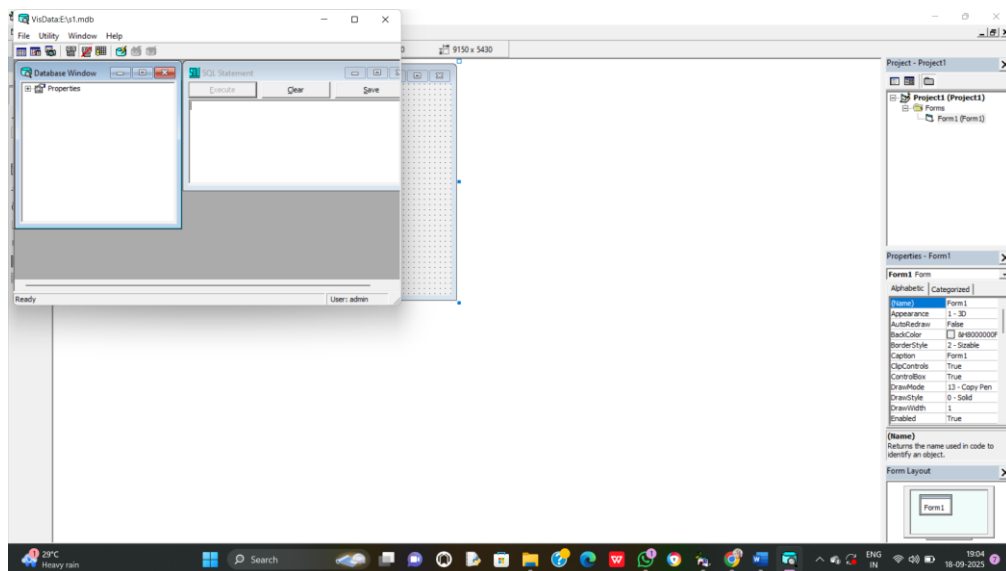
End Sub

Data Report

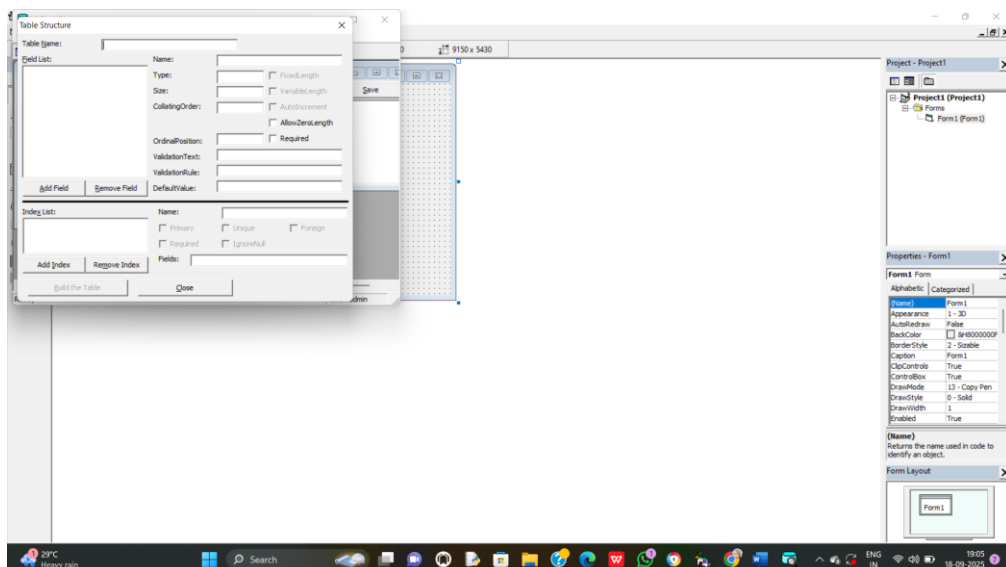
- ✓ In property window of data report1, Set the following
 - DataSource as “Data Environment1”
 - DataMember as “Command1”

How to Connect Visual Basic with Access database?

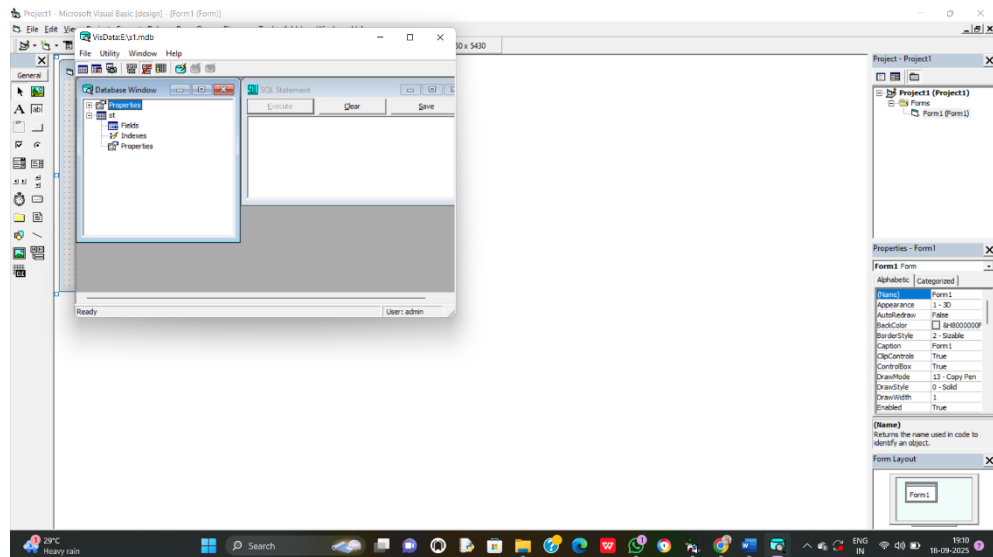
- 1) Design the application with desired control in VB 6.0
- 2) Goto **Add-Ins** in menu bar → select **Visual Data Manager**
- 3) Small VisData Window will appear
- 4) In that select **File → New → Microsoft Access → Version 7.0 MDB**
- 5) Choose the path and Type your database name like library.mdb and click “Save”



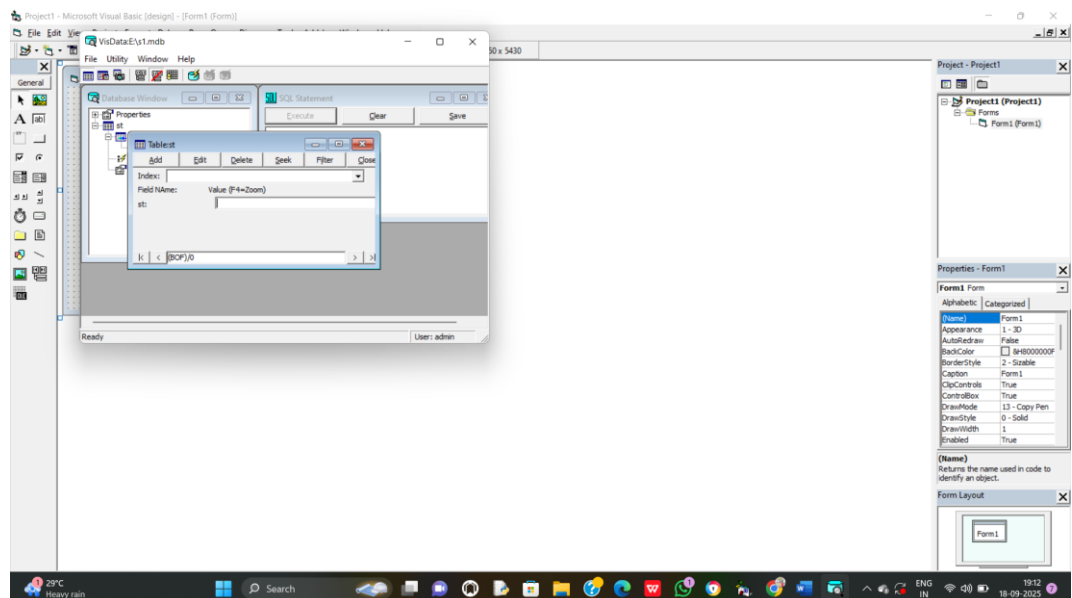
- 6) Right click on **Properties** and select “New Table”



- 7) Type the table name and click “Add Field”
- 8) Type the field name and select “type” of the field name
- 9) Once all the fields are added click “Ok” and then “Close”
- 10) Fields are added in the table and click “Build the Table”



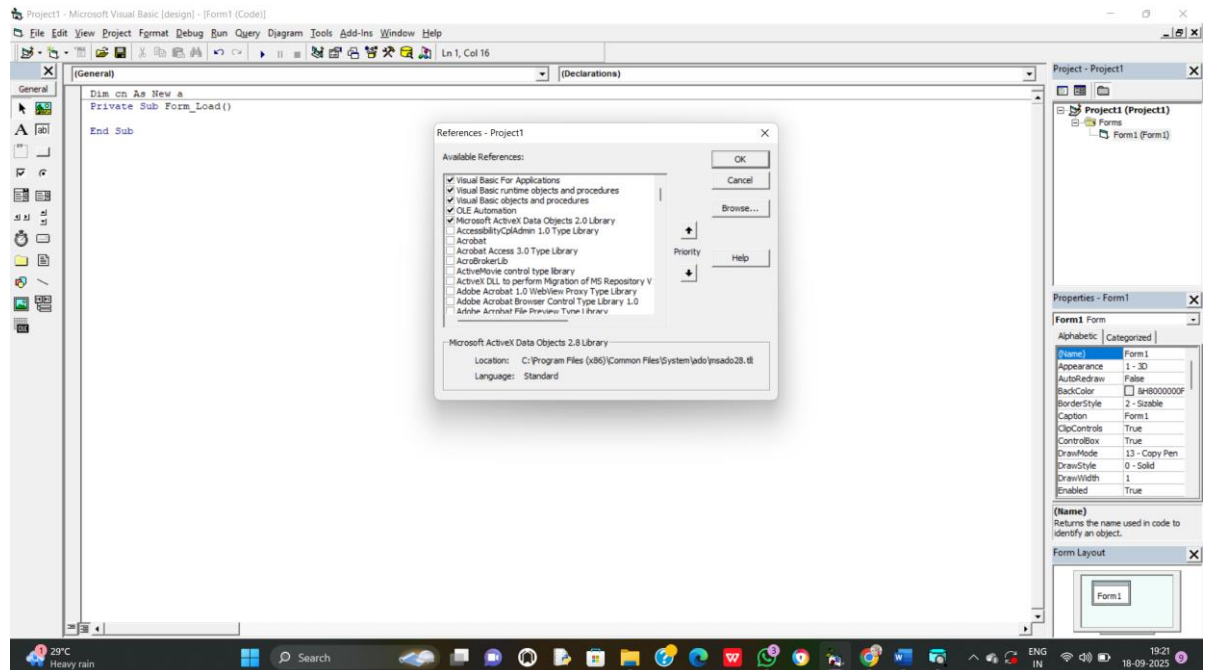
- 11) Right click on your table name and click “Open”



- 12) Click “Add”
- 13) Type the values for your field and click “Update”
- 14) That’s all for this close all the small window

If ADODB is Not added in your Project, proceed the following

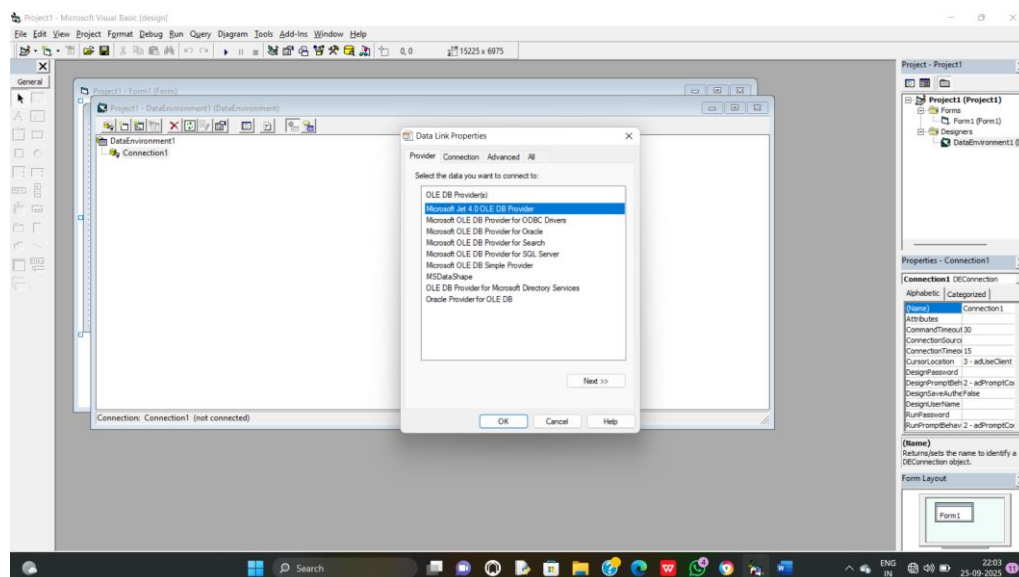
- 1) Click “Project” in Menu bar
- 2) Select “References”
- 3) In “Available References”, Tick the Checkbox “Microsoft ActiveX Data Object 2.0 Library”



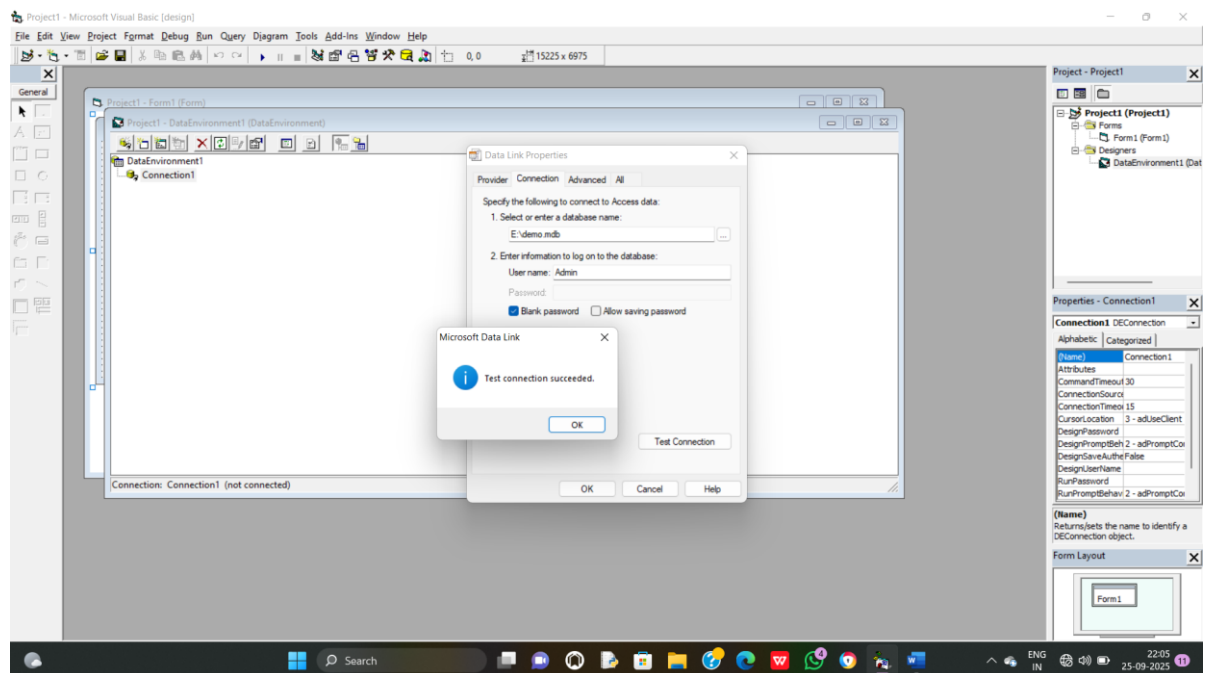
- 4) Then Click “Ok”

How to add Data Environment and Data Report?

1. In Project Explorer, Right click on Project1 → Add → Data Environment
2. Data Environment window will occur.
3. Right click on “Connection1” → Click Properties



4. Click “Microsoft Jet 4.0 OLE DB Provider” and then click “next”
5. Browse and select “Database Name” that was already created and click “Test Connection”



6. Click “OK”
7. Again, Right Click on “Connection1” → Add Command1
8. Right Click on “Command1” → Properties

Output:

Form1

LIBRARY MANAGEMENT SYSTEM

ADD BOOK

ISSUE BOOK

RETURN BOOK

VIEW REPORT

Windows taskbar: Search, 22:44, 17-09-2025

Form2

LIBRARY MANAGEMENT SYSTEM

ADD BOOK

BOOK ID: 1004

BOOK NAME: Computer Networks

AUTHOR'S NAME: Victor

QUANTITY: 23

ADD

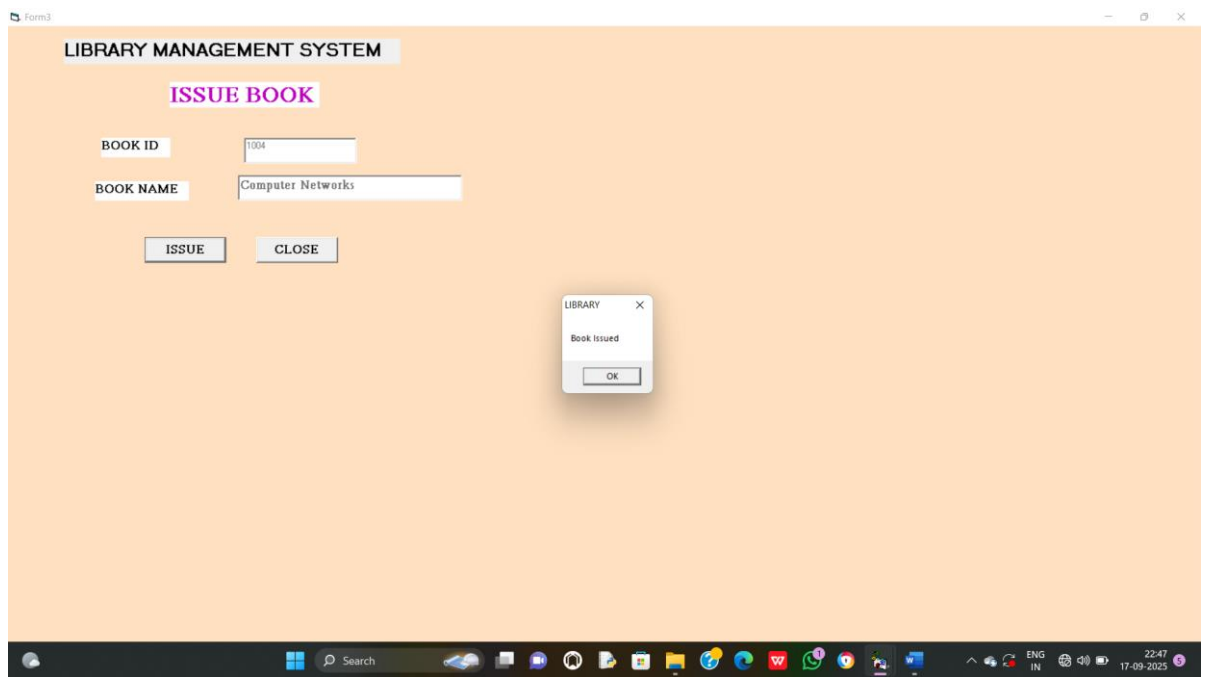
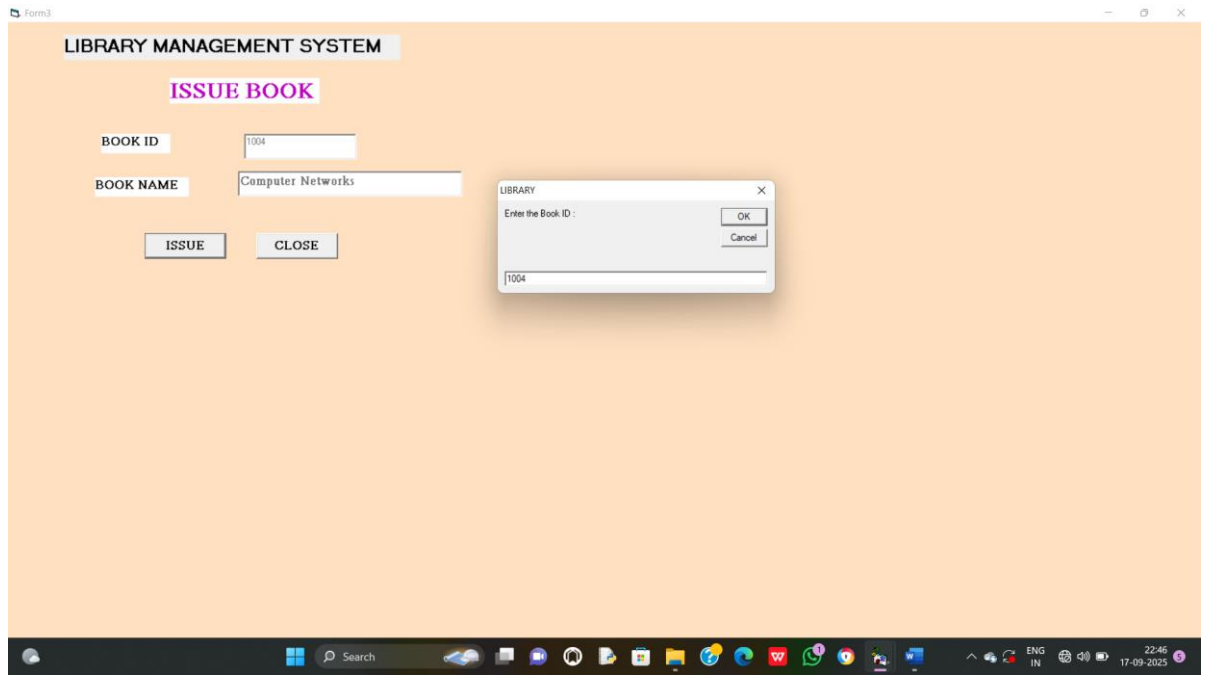
CLOSE

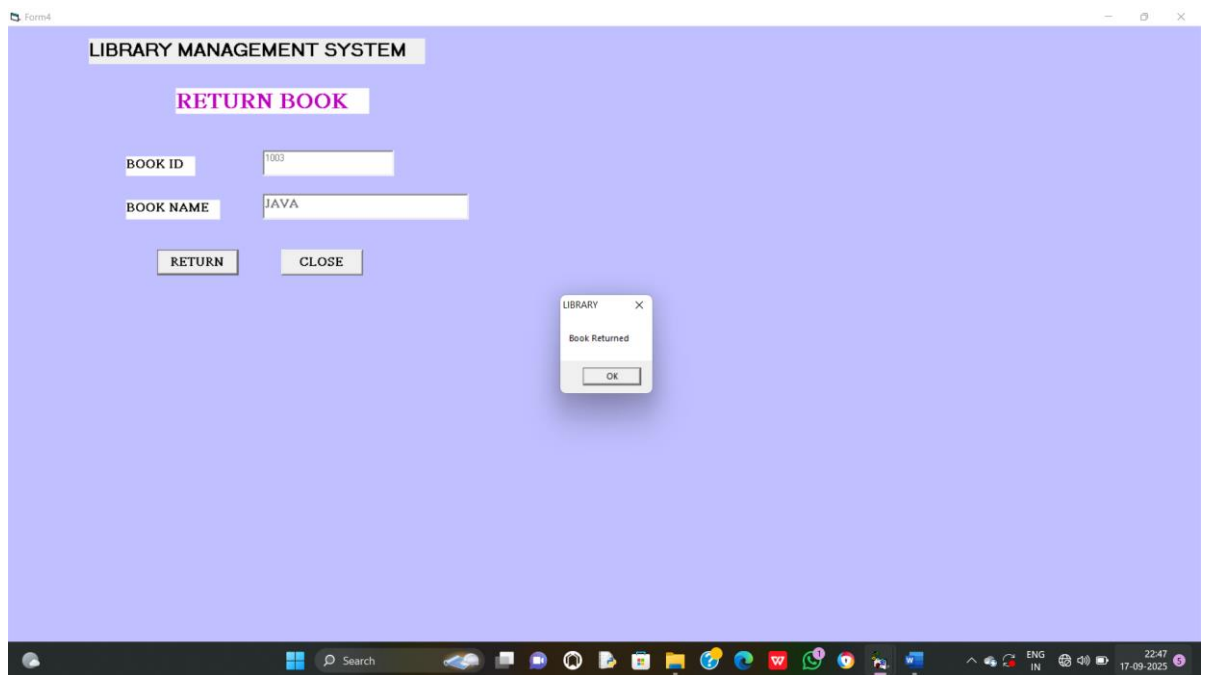
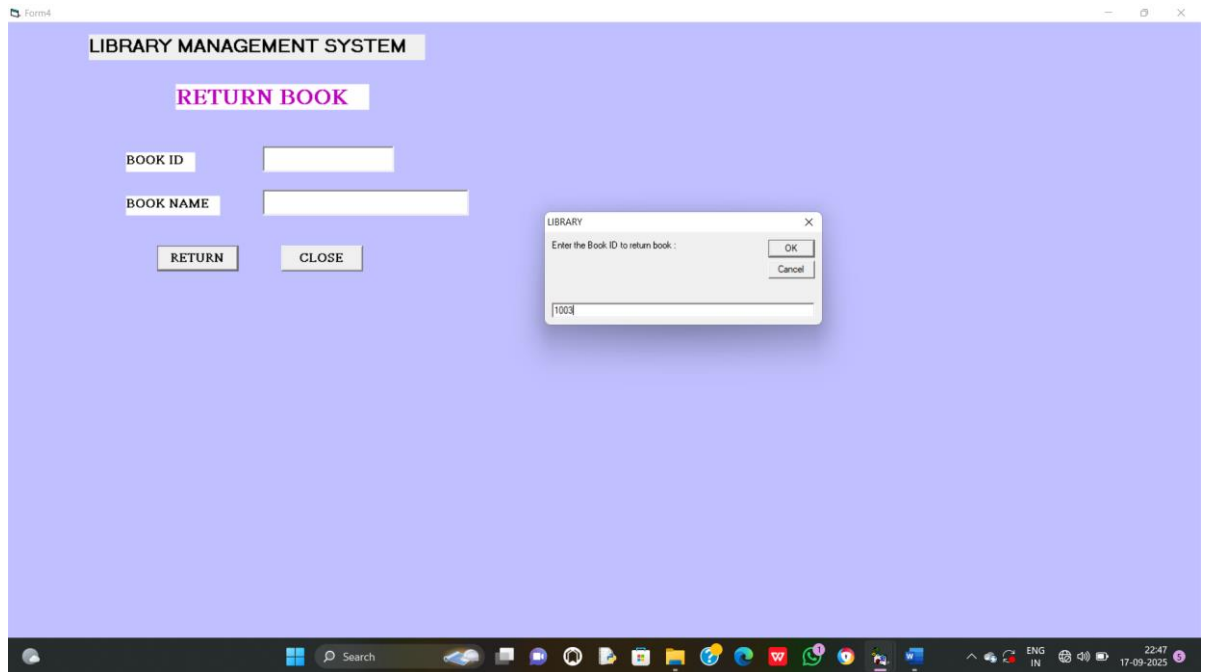
LIBRARY

Successfully inserted

OK

Windows taskbar: Search, 22:45, 17-09-2025





DataReport1

Zoom 100%

BOOK ID	BOOK NAME	AUTHOR'S	QUANTITY
1234	ROBMS	JOHN	20
1001	ad	dta	8
1003	JAVA	BALA GURUSAMY	17
1004	Computer Networks	Victor	21

Pages: 1/1

Search

ENG IN 22:48 17-09-2025

9) STUDENT MARKSHEET PROCESSING

Coding:

```
Dim cn As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

Private Sub calcmd_Click()

```
Dim tot As Integer
```

```
Text7.Text = Val(Text3.Text) + Val(Text4.Text) + Val(Text5.Text) + Val(Text6.Text)
```

```
If (Val(Text3.Text <= 30 And Text4.Text <= 30 And Text5.Text <= 30 And Text6.Text <= 30))
```

```
Then
```

```
    Text8.Text = "FAIL"
```

```
Else
```

```
    Text8.Text = "PASS"
```

```
End If
```

```
Text9.Text = (Val(Text7.Text) / 4)
```

```
End Sub
```

Private Sub delecmd_Click()

```
Dim studentID As String
```

```
Dim sql As String
```

```
studentID = InputBox("Enter the Student ID to delete:")
```

```
If Trim(studentID) = "" Then
```

```
    MsgBox "No ID entered. Operation cancelled."
```

```
    Exit Sub
```

```
End If
```

```
Set rs = New ADODB.Recordset
```

```
sql = "SELECT * FROM st WHERE std = " & studentID
```

```
rs.Open sql, cn, adOpenForwardOnly, adLockReadOnly
```

```
If rs.EOF Then
```

```
    MsgBox "No record found with ID: "
```

```
    rs.Close
```

```
    Exit Sub
```

```
End If
```

```
sql = "DELETE FROM st WHERE stid = " & studentID
```

```
cn.Execute sql
```

```
MsgBox "Record with ID " & studentID & " deleted successfully!"
```

```
End Sub
```

```
Private Sub Form_Load()
```

```
cn.Open "Provider=Microsoft.Jet.OLEDB.4.0;Data Source=E:\student.mdb;Persist Security Info=False"
```

```
rs.Open "select * from st", cn, adOpenKeyset, adLockOptimistic
```

```
End Sub
```

```
Private Sub insertcmd_Click()
```

```
rs.AddNew
```

```
rs.Fields(0).Value = Text1.Text
```

```
rs.Fields(1).Value = Text2.Text
```

```
rs.Fields(2).Value = Text3.Text
```

```
rs.Fields(3).Value = Text4.Text
```

```
rs.Fields(4).Value = Text5.Text
```

```
rs.Fields(5).Value = Text6.Text
```

```
rs.Fields(6).Value = Text7.Text
```

```
rs.Fields(7).Value = Text8.Text
```

```
rs.Fields(8).Value = Text9.Text
```

```
rs.Update
```

```
MsgBox "Succssfully Inserted"
```

Text1.Text = ""

Text2.Text = ""

Text3.Text = ""

Text4.Text = ""

Text5.Text = ""

Text6.Text = ""

Text7.Text = ""

Text8.Text = ""

Text9.Text = ""

End Sub

Private Sub reportcmd_Click()

DataReport1.Show

End Sub

Output:

Form1

STUDENT MARKSHEET PROCESSING

STUDENT ID	12345	CALCULATE	
STUDENT NAME	Madhu	TOTAL MARKS	324
OS	67	RESULT	PASS
RDBMS	89	PERCENTAGE	81
CN	90		
SE	78		

REPORT **INSERT** **DELETE**

student_db
Successfully Inserted
OK

Windows taskbar: Search, 19:25, 26-09-2025

Form1

STUDENT MARKSHEET PROCESSING

STUDENT ID		CALCULATE	
STUDENT NAME		TOTAL MARKS	
OS		RESULT	
RDBMS		PERCENTAGE	
CN			
SE			

REPORT **INSERT** **DELETE**

student_db
Enter the Student ID to delete:
234
OK Cancel

Windows taskbar: Rainy days ahead 29°C, Search, 19:40, 26-09-2025

Form1

STUDENT MARKSHEET PROCESSING

STUDENT ID		CALCULATE	
STUDENT NAME		TOTAL MARKS	
OS		RESULT	
RDBMS		PERCENTAGE	
CN			
SE			

REPORT INSERT DELETE

student_db

Record with ID 234 deleted successfully!

OK

Rainy days ahead
29°C

Search

ENG IN

19:41
26-09-2025

DataReport1

Zoom 100%

STUDENT ID	stname:	result:	percent:
67	dgh	PASS	69.75
34	dpdg	PASS	56.75
4545	fgsg	PASS	69.75

Pages: 1/1

Rainy days ahead
29°C

Search

ENG IN

19:41
26-09-2025

10) PAYROLL PROCESSING

Coding:

```
Dim cn As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

Private Sub calcmd_Click()

```
Text4.Text = Val(Text3.Text) * 0.25
```

```
Text5.Text = Val(Text3.Text) * 0.12
```

```
Text6.Text = Val(Text3.Text) + Val(Text4.Text)
```

```
Text7.Text = Val(Text6.Text) - Val(Text5.Text)
```

```
End Sub
```

Private Sub delcmd_Click()

```
Dim eid As String
```

```
Dim sql As String
```

```
eid = InputBox("Enter the EMPLOYEE ID to delete:")
```

```
If Trim(eid) = "" Then
```

```
    MsgBox "No ID entered. Operation cancelled."
```

```
    Exit Sub
```

```
End If
```

```
Set rs = New ADODB.Recordset
```

```
sql = "SELECT * FROM emp WHERE empid = " & eid
```

```
rs.Open sql, cn, adOpenForwardOnly, adLockReadOnly
```

```
If rs.EOF Then
```

```
    MsgBox "No record found with ID: "
```

```
    rs.Close
```

```
    Exit Sub
```

End If

sql = "DELETE FROM emp WHERE empid = " & eid

cn.Execute sql

MsgBox "Record with ID " & eid & " deleted successfully!"

End Sub

Private Sub exitcmd_Click()

End

End Sub

Private Sub Form_Load()

cn.Open "Provider=Microsoft.Jet.OLEDB.4.0;Data Source=E:\payroll.mdb;Persist Security Info=False"

rs.Open "select * from emp", cn, adOpenKeyset, adLockOptimistic

End Sub

Private Sub insertcmd_Click()

rs.AddNew

rs.Fields(0).Value = Text1.Text

rs.Fields(1).Value = Text2.Text

rs.Fields(2).Value = Text3.Text

rs.Fields(3).Value = Text6.Text

rs.Fields(4).Value = Text7.Text

rs.Update

MsgBox "Succssfully Inserted"

Text1.Text = ""

Text2.Text = ""

Text3.Text = ""

Text4.Text = ""

Text5.Text = ""

Text6.Text = ""

Text7.Text = ""

End Sub

Private Sub reportcmd_Click()

DataReport1.Show

End Sub

Data Report

- ✓ In property window of data report1, Set the following
 - DataSource as “Data Environment1”
 - DataMember as “Command1”

Table in MS Access

Database name : payroll

Table name : emp

FIELD NAME	TYPE
empid	Integer
empname	Text
salary	Double
gross	Double
net	Double

Output:

Form1

EMPLOYEE PAYROLL SYSTEM

EMPLOYEE ID	1001	INSERT DELETE DATA REPORT EXIT
EMPLOYEE NAME	Amudha	
SALARY	80000	
CALCULATE		
TOTAL ALLOWANCE	20000	
TOTAL DEDUCTION	9000	
GROSS SALARY	100000	
NET SALARY	90400	

PAYROLL
Successfully Inserted
OK

Windows taskbar: Search, 19:34, 19-09-2025

Form1

EMPLOYEE PAYROLL SYSTEM

EMPLOYEE ID		INSERT DELETE DATA REPORT EXIT
EMPLOYEE NAME		
SALARY		
CALCULATE		
TOTAL ALLOWANCE		
TOTAL DEDUCTION		
GROSS SALARY		
NET SALARY		

PAYROLL
Enter the EMPLOYEE ID to delete:
1237
OK
Cancel

Windows taskbar: Search, 19:34, 19-09-2025

Form1

EMPLOYEE PAYROLL SYSTEM

EMPLOYEE ID

EMPLOYEE NAME

SALARY

CALCULATE

TOTAL ALLOWANCE

TOTAL DEDUCTION

GROSS SALARY

NET SALARY

INSERT

DELETE

DATA REPORT

EXIT

PAYROLL

Record with ID 1237 deleted successfully!

OK

Windows taskbar: Search, File Explorer, Microsoft Edge, Word, Teams, OneDrive, Outlook, Settings, ENG IN, 19:34 19-09-2025

DataReport1

Zoom 100%

EMPID	NAME	SALARY	GROSS SAL	NET SALARY
1234	HARI	80000	100000	90400
1235	ABI	78000	97500	88140
1001	Amudha	80000	100000	90400

Pages: 1 | 1

Windows taskbar: Search, File Explorer, Microsoft Edge, Word, Teams, OneDrive, Outlook, Settings, ENG IN, 19:35 19-09-2025